

Anhad Mehrotra

+91 9311061194 | anhad01mehrotra@gmail.com | anhadmehrotra.dev |
anhad-mehrotra | Anhad-01 | New Delhi, India

EDUCATION

Cluster Innovation Centre, University of Delhi

Nov. 2022 – Present

B.Tech in Information Technology and Mathematical Innovations

New Delhi, India

- **CGPA:** 9.11 / 10
- **Coursework:** Artificial Intelligence, Computer Vision, Data Structures, Probability & Statistics, Linear Algebra

SKILLS

Languages : Python, C, Java, SQL

ML / DL : TensorFlow, Keras, PyTorch, scikit-learn, OpenCV, NLTK, FAISS, SentenceTransformers, LangChain, CrewAI

Data : NumPy, Pandas, SciPy, Power BI

Tools : Git, Docker, FastAPI, Linux, MATLAB, Arduino, COMSOL Multiphysics, Jupyter, Colab

Web : HTML, CSS, React.js, Streamlit, WordPress

EXPERIENCE

Edxso

Feb. 2026 – Apr. 2026

AI Engineer Intern

New Delhi, India

- Built an AI assessment platform evaluating question papers against competency frameworks (NCF), generating alignment reports with RBAC.
- developed a GenAI pipeline for automated product visual creation supporting single and batch workflows.
- Contributed to a scalable question bank system scoring submissions against defined competencies with a reward mechanism to incentivize quality contributions.

National e-Governance Division, Ministry of Electronics & IT

Jun. 2025 – Aug. 2025

AI/ML Intern

New Delhi, India

- Built an agentic AI-powered course recommender using hybrid filtering and semantic search.
- Researched AI security threats, governance frameworks, and operational measures including API security and red teaming.
- Developed multilingual semantic search and feature enhancements for the myscheme portal and mobile app.

Craving For Gaming

Sep. 2024 – Feb. 2025

Web Developer Intern

New Delhi, India

- Optimized website performance and improved UX using HTML/CSS and on-page SEO techniques.
- Reduced page load time and improved keyword ranking across 5+ product segments.
- Supported a product listing revamp that **improved user engagement by 18%**.

PROJECTS

Nanoparticle Toxicity Prediction Using Deep Learning

- Designed a custom deep neural network for cross-dataset nanotoxicity prediction; benchmarked against a Random Forest baseline on a general nanotoxicity dataset.
- **Improved cross-dataset accuracy from 75% (RF) to 86.75%** on a lung macrophage dataset.
- Applied PDP, 2D-PDP, and SHAP analyses to identify dominant toxicity factors.
- **GitHub:** Nanotoxicity Prediction

Vision-Language Fusion for Image Captioning

- Built a hybrid captioning pipeline combining YOLOv5, Swin Transformer, and Mask R-CNN with enhanced Weighted Box Fusion (WBF); achieved **83.2% accuracy on Pascal-50S** with entity-aware structured captioning.
- Deployed a real-time captioning dashboard via Streamlit.
- **GitHub:** [Image Captioning](#)

IoT-based Air & Sound Monitoring and Forecasting

- Developed an ESP32-based system to collect and forecast environmental data using MQ-135, DHT11, and LM393 sensors deployed in Delhi.
- Used K-Means for clustering and LSTM for time-series pollution prediction.
- **GitHub:** [Air and Sound Monitoring and Prediction](#)

ACHIEVEMENTS

TATA Technologies InnoVent Hackathon | *Top 31 / 2,800+ teams*

- Built an AI-driven EV battery analytics platform with transformer- and RNN-based models for RUL estimation, achieving **85%+ accuracy**; dual-layer architecture for data processing and API delivery. [GitHub](#)

Hack The Future Hackathon, MoSPI | *Top 20 / 1,000+ teams*

- Built a multilingual semantic search app for NIC code classification with voice-to-text and synonym mapping; **improved relevance by 21%** over baseline. [GitHub](#)

PUBLICATIONS

Cryosurgical Treatment Planning for Brain Tumor via CNN Segmentation | *Cryobiology, Elsevier*

- U-Net MRI segmentation (**99.74% accuracy**, 0.872 Dice) integrated with K-Medoids, GMM, and BPA optimization for cryoprobe placement; validated with COMSOL Multiphysics thermal simulation. [DOI](#)